SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Assessment Detail	Information	Assessment Detail	Information
Course Code	CSA10 / CSA1021	Course Title	Software Engineering
CO Assessed	CO1	Assessment	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage	20%	Total Marks	25
Duration	60 minutes	Date	14/10/2026
Student Name	P. Pramod Kumar	Reg. No.	1925260245

CO1 Statement

Apply advanced methodologies and agile project management techniques to manage software projects effectively.

Project Topic

Decentralized Lifelong Learning Ledger & Micro-Credential Verification Matrix

Code of Conduct

I, P. Pramod Kumar (1925260245), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P.Pramod Kumar



Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students

Based on the assigned problem statement, apply Agile software engineering practices and prepare a sprint planning document covering the following assessment questions.

1. **Identify the Project Vision and Stakeholders**
   Define the project vision, objectives, and key stakeholders involved.

2. **Create Epics and User Stories**
   Break down requirements into epics and write stories using: As a <user>, I want <feature>, so that <benefit>.

3. **Define Acceptance Criteria**
   Write clear and measurable acceptance criteria for each selected story using Given–When–Then or an equivalent format.

4. **Prioritize the Product Backlog**
   Order backlog items based on business value, user needs, risk, and dependencies.

5. **Plan the Sprint Backlog**
   Select high-priority stories for Sprint 1 and break them into actionable development tasks.

6. **Estimate Effort Using Story Points**
   Assign story points using Fibonacci values or Planning Poker and justify each estimate.

7. **Present Sprint Goal and Expected Deliverables**
   Define the Sprint Goal and the deliverables expected at the end of the sprint.

Project context supplied with the assignment: job applicants collect credentials from courses, bootcamps, and workplace training; employers need fast verification. The proposed platform provides a secure lifelong learning registry and shareable digital portfolio.

Assessment Rubrics for Calculation

The following rubric is reproduced in a readable format from the assessment sheet.

Criteria	Excellent (3)	Good (2)	Satisfactory (2)	Needs Improvement (1)
Project Vision & Stakeholder Identification (30%)	Clear, comprehensive vision; all relevant stakeholders and roles identified.	Clear vision with minor omissions; most stakeholders identified.	Partially defined vision; some stakeholders missing or incorrect.	Unclear vision; stakeholders poorly identified or missing.
Epics & User Stories (25%)	Well-defined epics and complete stories in Agile format capturing requirements.	Mostly complete with minor formatting or requirement issues.	Partially complete with limited Agile structure.	Incomplete or unclear stories; Agile format not followed.
Acceptance Criteria (20%)	Specific, measurable, testable and aligned with stories.	Mostly clear and testable with minor gaps.	Partially defined and lacking clarity/completeness.	Missing or not aligned with stories.
Backlog Prioritization & Sprint Planning (15%)	Logical priority, realistic sprint backlog and complete task breakdown.	Mostly appropriate with minor issues.	Partially complete; limited justification.	Poorly organized or incomplete.
Estimation, Sprint Goal & Presentation (10%)	Well-justified estimates, clear goal, realistic deliverables and professional presentation.	Mostly appropriate with minor inconsistencies.	Some justification or deliverables incomplete.	Unrealistic/ missing estimates; unclear goal or presentation.

Scoring Item	Marks
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

Problem Statement and Project Background

Industry Context

Job markets are moving away from relying only on traditional four-year degrees and increasingly value continuous skills development and micro-credentials. Applicants earn certifications through online courses, bootcamps, universities, and workplace training, but these records are often scattered across different providers. Applicants may struggle to compile, share, and verify their achievements, while employers face slow and expensive manual verification. A secure lifelong learning registry is needed so people can present their skills and allow authorized verifiers to check credentials quickly.

Problem Statement

Design and develop a decentralized Lifelong Learning Ledger that allows educational institutions, employers, and online learning platforms to issue tamper-evident micro-credentials. The platform must maintain a secure registry of skill badges, course completions, and professional certifications. Learners should be able to organize these milestones into a digital portfolio and share selected credentials with recruiters for rapid verification without depending on a central school registrar.

Project Aim

To develop a secure, privacy-conscious and decentralized digital registry that enables learners to collect, manage, share, and verify lifelong learning credentials from multiple trusted providers, while reducing the time and cost of manual credential checks.

Objectives

Build a decentralized platform for issuing, claiming, and organizing digital skill credentials.

Implement cryptographic validation to help detect document tampering and forged credentials.

Provide secure, consent-based sharing so learners control who can view selected credentials.

Support integration with HR and recruitment tools to simplify skill verification during job applications.

Allow credentials to be updated, corrected, expired, or revoked through controlled status management.

Support evolving micro-credentials as technologies, skills, and job roles change over time.

Expected Outcomes

Faster and lower-cost verification of professional skills for hiring managers and recruiters.

A portable, verified portfolio that can improve career mobility across employers and learning providers.

Reduced administrative workload for schools, training academies, and online course providers through more automated issuing.

A clearer, data-informed view of skills and learning achievements, while protecting personal information.

1. Project Vision and Problem Understanding

Vision: Create a privacy-conscious, decentralized lifelong learning registry that lets learners collect verified micro-credentials in one portfolio and lets an authorized recruiter confirm their authenticity quickly, without depending on a single school registrar.

Problem summary: Skills are earned across universities, online learning platforms, bootcamps, and workplace training. Records are scattered across providers and may be difficult to verify. Manual checks increase recruitment time and administrative cost. The solution should make credential proofs tamper-evident, portable, and selectively shareable.

Project objectives

- Allow trusted issuers to create and digitally sign micro-credentials.
- Give learners a consolidated portfolio of course completions, skill badges, and professional certificates.
- Enable consent-based sharing with recruiters or other verifiers.
- Validate issuer authenticity, credential integrity, and current status.
- Support revocation or correction of credentials without rewriting the learner's entire portfolio.
- Reduce repetitive manual verification while protecting personal information.

Scope and boundaries

The first prototype focuses on account roles, credential metadata, issuing, portfolio viewing, share links, and verification status. It does not claim to replace formal accreditation, prove a learner's real-world competence by itself, or make hiring decisions. The ledger records proofs and references; personal details and full certificate files should remain off-chain.

2. Stakeholder Identification and Responsibilities

Stakeholder	Needs / concerns	Responsibilities in the solution
Learner / credential holder	Portable portfolio, privacy, simple sharing, correction process.	Claim credentials, organize portfolio, grant/revoke sharing consent.
Educational institution / issuer	Trusted issuance, fewer manual requests, audit trail.	Verify learner identity, issue signed credentials, correct/revoke records.
Online course / bootcamp provider	Fast issuing and integration, clear credential format.	Provide course completion evidence and issuer identity.
Employer / recruiter	Reliable, quick, easy-to-read verification.	Check validity and relevance; avoid requesting unnecessary personal data.
Platform administrator	Availability, security, incident response.	Manage service configuration and access roles; cannot silently alter issuer signatures.
Development team	Clear requirements and testable acceptance criteria.	Design, build, test, document, and demonstrate each increment.
Privacy / compliance reviewer	Consent, retention limits, data minimization.	Review privacy notices, access logs, and applicable legal obligations.
External verifier / auditor	Traceability and evidence of issuer status.	Review verification evidence and audit records where authorized.

Stakeholder engagement plan

- Before each sprint: Product Owner confirms priority and acceptance criteria with learner and recruiter representatives.
- During development: daily short team sync identifies blockers and updates task status.
- At Sprint Review: demonstrate a working issue–share–verify path and collect feedback.
- At Retrospective: identify process improvements, including test coverage, handoffs, and privacy review.

3. Functional and Non-Functional Requirements

Functional requirements

1. FR1 – Role access: the system distinguishes learners, credential issuers, verifiers, and administrators.
2. FR2 – Issuer onboarding: an issuer account is verified and its public verification key or trusted identity is registered.
3. FR3 – Credential issuance: an authorized issuer creates a credential with learner reference, title, skills, issue date, issuer ID, and unique credential ID.
4. FR4 – Integrity proof: the platform calculates a cryptographic hash and verifies the issuer's digital signature.
5. FR5 – Portfolio: learners view and organize credentials in a chronological, searchable portfolio.
6. FR6 – Consent-based sharing: a learner creates a limited link or QR code for selected credentials and can expire or revoke access.
7. FR7 – Verification: a recruiter checks issuer, signature/hash, expiry if applicable, and revocation status.
8. FR8 – Status handling: a credential can be active, expired, revoked, or under review.
9. FR9 – Audit events: issuance, verification, and sharing actions are logged with minimal necessary metadata.
10. FR10 – Feedback: users receive clear success, invalid, expired, or unavailable messages.

Non-functional requirements

Quality area	Target for prototype	How to verify
Security	Role-based access; signed credentials; secure session handling.	Negative access tests and signature tampering tests.
Privacy	Keep personal data and certificate files off-chain; share only selected fields.	Inspect database and share-link payloads.
Performance	Typical verification should complete within 3 seconds in the demo environment.	Run 30 sample verifications and record response time.
Availability	Graceful error when ledger or issuer service is unavailable.	Simulate unavailable dependency and check error message.
Usability	A first-time recruiter can verify a sample credential without training.	Observe 3 users completing a test scenario.
Maintainability	Documented API, modular issuer/portfolio/verifier components.	Code review and setup test by another developer.

4. Proposed Solution Architecture and Feasibility

The proposed system uses a web application, an API layer, an issuer identity registry, a credential metadata store, a ledger or append-only proof store, and a verification service. The learner's private details and certificate file remain in protected off-chain storage. The ledger stores a hash, issuer reference, timestamp, and status pointer, not the full personal record.

Architecture components

- User interface: learner dashboard, issuer console, recruiter verification page.
- Authentication and authorization: role-based access and consent checks.
- Credential service: validates required fields, creates a unique ID, computes a hash, and applies issuer signature.
- Proof registry: stores the proof or transaction reference so later tampering can be detected.
- Verification service: recalculates the hash, checks signature and issuer status, and checks revocation state.
- Audit and monitoring: records security-relevant events without exposing unnecessary personal data.

Feasibility analysis

Feasibility	Assessment	Reason and condition
Technical	Feasible for a prototype	Standard web frameworks, cryptographic libraries, signed JSON records, and a permissioned/test ledger can demonstrate the flow. Production-grade identity assurance needs more design.
Operational	Feasible with issuer onboarding	Institutions must agree on credential fields, issuer identity checks, correction and revocation procedures.
Economic	Moderate / manageable	A prototype can use open-source tools and a test network. Production costs include hosting, security review, integrations, and support.
Schedule	Feasible in short sprints	A narrow first sprint can deliver issuer registration, sample credential issuance, and basic verification; advanced integrations follow later.
Legal and privacy	Feasible only with safeguards	Applicable privacy, retention, accessibility, and education-record rules must be reviewed. Immutable storage must not contain personal data.

Adoption	Requires collaboration	Recruiters need a simple verification page and issuers need incentives and low-effort integration.

5. Comparison of Possible Approaches

Criterion	Centralized database	PDF email verification	Proposed decentralized proof ledger
Integrity / tamper evidence	Depends on administrator controls and audit logs.	Files can be edited or forged; manual checks needed.	Hash and issuer signature make record alteration detectable.
Single point of control	Usually one platform operator.	No shared source of truth.	Proof can be checked against a shared registry, depending on network design.
Verification speed	Fast if all providers integrate; otherwise variable.	Often slow and manual.	Fast for a supported credential once issuer trust and status are available.
Privacy	Central operator holds records; access policy is critical.	Documents may be widely forwarded.	Selective sharing possible; private data should remain off-chain.
Revocation / correction	Can update central record quickly.	Recipients may retain outdated copies.	Status registry can mark revoked; clients must check status each time.
Cost / complexity	Lower initial complexity; central operations needed.	Low setup cost but high manual labour.	Higher initial integration and governance complexity.
Best fit	One organization or tightly integrated ecosystem.	Small volume and informal verification.	Multiple issuers needing portable, independently verifiable proofs.

Decision: use a hybrid approach. Keep personal records in a controlled off-chain store and publish only a cryptographic proof or reference to a ledger. This avoids treating blockchain as a place to store all data and gives a practical route for privacy and revocation.

6. Epics and Product Backlog Overview

Epic ID	Epic	Business value	Key capabilities
E1	Identity and role access	High	Sign-in, role permissions, issuer trust setup.
E2	Credential issuance	Very high	Credential fields, unique ID, signature, proof creation.
E3	Learner portfolio	High	View, search, sort, and group credentials.
E4	Selective sharing	High	Consent-based share link, expiry, revoke access.
E5	Credential verification	Very high	Check issuer, signature/hash, and current status.
E6	Revocation and corrections	High	Status updates, reason, audit trail.
E7	Audit and reporting	Medium	Event logs and aggregate usage metrics.
E8	Integrations and accessibility	Medium	Issuer APIs, HR workflow, accessible interface.

Definition of Ready

- User story has a clear user and benefit.
- Acceptance criteria are testable.
- Dependencies and data fields are identified.
- The team can estimate the work.
- Privacy and security implications are noted.

Definition of Done

- Code is reviewed and merged.
- Unit and integration tests pass.
- Acceptance criteria are demonstrated.
- No known critical security defect remains in the tested scope.
- User-facing errors and basic documentation are present.
- The increment is deployable to the test environment.

7. User Stories – Issuance and Identity

ID / Priority	User story	Acceptance criteria (Given/When/Then)
US-01 / Must	As an issuer, I want to register my organization identity so that learners and recruiters can recognize trusted credentials.	Given an issuer submits required details, when an authorized reviewer approves them, then the issuer status is shown as trusted; unapproved issuers cannot issue production credentials.
US-02 / Must	As an authorized issuer, I want to issue a signed micro-credential so that a learner can prove course completion.	Given valid required fields and an active issuer, when issue is selected, then a unique ID, timestamp, signature, and proof reference are created.
US-03 / Must	As a learner, I want to claim a credential so that it appears in my portfolio.	Given a valid claim token, when the learner accepts it, then the credential is linked to the account and the token cannot be reused.
US-04 / Should	As an issuer, I want to correct or revoke a credential so that inaccurate records are not treated as valid.	Given an authorized issuer submits a reason, when the change is approved, then status changes and an audit event is recorded.

Detailed Given–When–Then example: US-02

Given an active trusted issuer is signed in and all required credential fields are present, when the issuer submits the credential, then the service validates the fields, generates a unique credential ID, computes a hash, signs the credential, stores the proof reference, and displays a success receipt. If required fields are missing, no credential is issued and a field-level error is shown.

8. User Stories – Portfolio, Sharing and Verification

ID / Priority	User story	Acceptance criteria (Gherkin)
US-05 / Must	As a learner, I want to see all claimed credentials in one portfolio so that I can present my lifelong learning history.	Given the learner has credentials, when the portfolio opens, then each item shows title, issuer, issue date, skills, and current status.
US-06 / Must	As a learner, I want to share selected credentials with a recruiter so that I control what is disclosed.	Given credentials are selected, when the learner creates a share link, then only selected fields are visible and expiry/disable controls are available.
US-07 / Must	As a recruiter, I want to verify a credential so that I can confirm authenticity without contacting the school manually.	Given a valid share link or credential ID, when the recruiter verifies it, then issuer, signature/hash, and status are checked and a clear result is returned.
US-08 / Should	As a learner, I want to revoke a share link so that access ends when it is no longer needed.	Given a share link is active, when the learner revokes it, then future requests return access denied.
US-09 / Could	As an administrator, I want aggregate usage reports so that I can monitor service reliability.	Given authorized access, when a report is requested, then aggregate counts are shown without exposing unnecessary learner data.

Verification result labels

- VALID – proof, issuer, and status checks pass.
- INVALID – signature/hash mismatch or unknown credential.
- REVOKED – issuer has marked the credential revoked.
- EXPIRED – the credential has a defined expiry that has passed.
- UNAVAILABLE – verification dependency could not be reached; do not label it valid.

9. Acceptance Criteria and Test Scenarios

Test ID	Scenario	Expected result
AT-01	Issue credential with all valid fields.	Credential ID and proof reference created; success receipt displayed.
AT-02	Attempt issuance by an untrusted issuer.	Request rejected; no proof record created.
AT-03	Alter one character in a stored credential after issuance.	Hash/signature validation fails; result is INVALID.
AT-04	Verify a valid, active credential.	Result is VALID with issuer and issue date shown.
AT-05	Verify a revoked credential.	Result is REVOKED; no green valid indicator.
AT-06	Use an expired share link.	Access is denied and the user is prompted to request a new link.
AT-07	Learner shares only one selected credential.	Recruiter sees only the selected credential and allowed fields.
AT-08	Verification service is unavailable.	System shows UNAVAILABLE and does not falsely claim authenticity.
AT-09	Learner revokes a share link.	Subsequent access attempts fail.
AT-10	Submit incomplete credential fields.	Field-level validation message; no credential is issued.

Acceptance test approach

Use a combination of unit tests for hashing/signatures, API integration tests for issue and verify operations, role-permission tests, UI tests for sharing, and manual usability checks with a learner and recruiter. Test both successful and rejected cases; security failures must fail closed.

10. Product Backlog Prioritization

Priority uses MoSCoW (Must, Should, Could, Won't for the first release) with dependency and risk considered. The order is a planning proposal for a student prototype, not measured production data.

Rank	Backlog item	Priority	Value / dependency / rationale
1	Issuer identity and trusted status	Must	Verification is meaningful only if the issuer can be trusted.
2	Credential data model and issuance	Must	Core product capability; creates the record to verify.
3	Hash and signature verification	Must	Core integrity check; depends on issuance format.
4	Learner portfolio view	Must	Allows learners to collect and inspect records.
5	Recruiter verification page	Must	Demonstrates the principal user benefit.
6	Revocation status check	Must	Prevents revoked credentials being accepted as current.
7	Consent-based share links	Must	Enables practical, privacy-aware recruiter access.
8	Issuer API integration	Should	Reduces manual issuing but needs stable data contracts.
9	Analytics dashboard	Could	Useful for operations but not required to prove the core flow.
10	Multi-ledger interoperability	Won't for MVP	High complexity; defer until data model and trust governance are proven.

Why this order is realistic

The first release should prove one end-to-end path before adding analytics or integrations: a trusted issuer creates a signed credential; the learner claims and shares it; a recruiter checks its proof and status. This reduces the risk of building dashboards around a verification process that has not yet been validated.

13. Story Point Estimation and Rationale

Use Planning Poker with a modified Fibonacci scale (1, 2, 3, 5, 8, 13). Story points represent relative effort, complexity, and uncertainty—not hours. The estimates below are initial team forecasts and should be revised after discussion.

Story	Points	Estimate rationale	Main uncertainty
US-01 issuer trust	3	Small workflow with validation and role status.	How issuer approval is represented.
US-02 issue credential	5	Multiple fields, unique ID, hashing, signing, and error handling.	Cryptographic library integration and key handling.
US-05 portfolio	3	Read-only list and basic credential details.	Account-to-credential linking.
US-06 share selected credentials	5	Consent, selected fields, token expiry, and access control.	Secure token design and privacy tests.
US-07 verify credential	5	Recompute hash, validate signature, check issuer and status.	Status registry availability and error states.
US-08 revoke link	3	Link status update and denial after revocation.	Cache invalidation and edge cases.
US-04 revoke credential	5	Issuer authorization, status update, reason, audit event.	Ledger/status design and policy rules.

Estimation method

11. Read the story and acceptance criteria together.
12. Each team member privately selects a point value.
13. Reveal estimates simultaneously and discuss the largest differences.
14. Clarify hidden work, dependencies, and test requirements.
15. Vote again until a reasonable shared estimate is reached.
16. Compare completed work at the retrospective; do not treat one sprint as a permanent velocity.

15. Security, Privacy and Data Governance

Security controls

- Use HTTPS for all network communication and secure password/session practices.
- Apply least-privilege role permissions to issuing, revoking, viewing, and administration.
- Use standard, reviewed cryptographic libraries rather than custom cryptography.
- Protect issuer private keys; keep private keys out of source control and logs.
- Validate input fields and rate-limit public verification endpoints.
- Log security-relevant actions without storing unnecessary sensitive information.

Privacy-by-design controls

- Store personal details and full certificate documents off-chain in access-controlled storage.
- Publish only the minimum proof data needed for integrity verification.
- Use short-lived, revocable share links and avoid exposing learner identity in URLs.
- Provide clear consent language describing exactly what a recruiter can view.
- Define retention and deletion rules for off-chain data and logs.
- Review applicable Indian data protection and institutional policies before real deployment.

Threat model summary

Threat	Control	Residual issue
Credential alteration	Hash comparison and digital signature verification.	A valid signature does not prove the issuer's original claim was truthful.
Impersonated issuer	Issuer onboarding and trusted-key registry.	Weak onboarding can still admit a fraudulent issuer.
Unauthorized portfolio access	Authentication, authorization, short-lived share tokens.	Compromised user account remains a risk.
Link forwarding	Limited scope, expiry, revoke function.	A recipient may copy information already viewed.
Unavailable status service	Explicit UNAVAILABLE response; fail closed.	Temporary inability to verify can delay hiring.

20. Conclusion and References

Conclusion

The Decentralized Lifelong Learning Ledger addresses a practical problem: learners need a portable record of skills earned from many providers, while employers need trustworthy verification without repeated manual contact. A hybrid design—signed credentials, cryptographic integrity proofs, off-chain personal data, learner-controlled sharing, and live status checks—offers a feasible path for a small prototype. Agile planning makes the work manageable by prioritizing issuer trust, issuance, and verification before advanced integrations. The proposed first sprint has a focused goal, measurable acceptance criteria, a 16-point illustrative forecast, and explicit negative tests. Production readiness would require stronger identity assurance, key management, privacy/legal review, accessibility validation, and participation from real issuing organizations.

References

1. **Schwaber, K., & Sutherland, J. (2020). The Scrum Guide. https://scrumguides.org/scrum-guide.html**
2. **Scrum Guides. (2020). The 2020 Scrum Guide – official download page. https://scrumguides.org/download.html**
3. **National Institute of Standards and Technology (NIST). (2018). Digital Identity Guidelines, SP 800-63-3. https://pages.nist.gov/800-63-3/**
4. **W3C. (2022). Verifiable Credentials Data Model v1.1. https://www.w3.org/TR/vc-data-model/**
5. **W3C. (2023). Verifiable Credentials Data Model v2.0 (Working Group / standards-track materials). https://www.w3.org/TR/vc-data-model-2.0/**
6. **OWASP Foundation. Application Security Verification Standard (ASVS). https://owasp.org/www-project-application-security-verification-standard/**
7. **Assignment source: CSA10 / CSA1021 CO1 Assessment Tool 2 – Agile Sprint Planning & User Story Workshop, supplied PDF (2 pages), including the problem statement screenshots supplied with this task.**

Note: The Agile assessment instructions and rubric are taken from the supplied assignment sheet. Technical architecture, user stories, estimates, test scenarios, and feasibility assessments are proposed design work for the stated project, not reported results from a deployed system.

Jira Software – Sprint Board

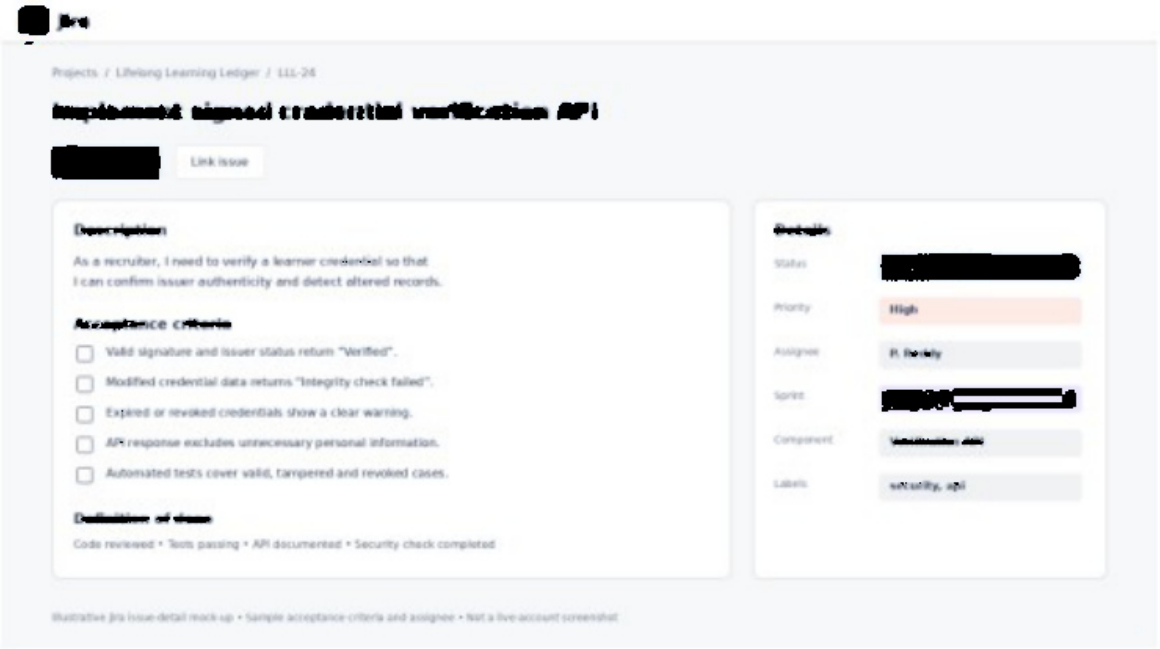
Use a Kanban-style board to monitor Sprint 1 tasks for the Decentralized Lifelong Learning Ledger. Suggested workflow: To Do → In Progress → Review → Testing → Done.



Each issue should identify an owner, estimate, acceptance criteria, dependency, and blocker. Review progress daily against the Sprint Goal.

Jira Software – Issue Details

Example issue: US-07 – Verify credential. Include testable conditions for issuer trust, signature/hash integrity, and current status.



Expected outcomes distinguish VALID, INVALID, REVOKED, EXPIRED, and UNAVAILABLE. These are design mock-ups, not evidence of a live Jira project or executed tests.